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CHAPTER 1 • NUMBER • TOPIC 2

Sets & Venn Diagrams

Learn four symbols, then every question is listing, counting or filling regions — always from the middle out.



$$\frac{3}{10} = 0.3 =$$

30%

6

VARIANTS

25

PAST-PAPER PARTS

32

PRACTICE QUESTIONS

YOU WILL MASTER

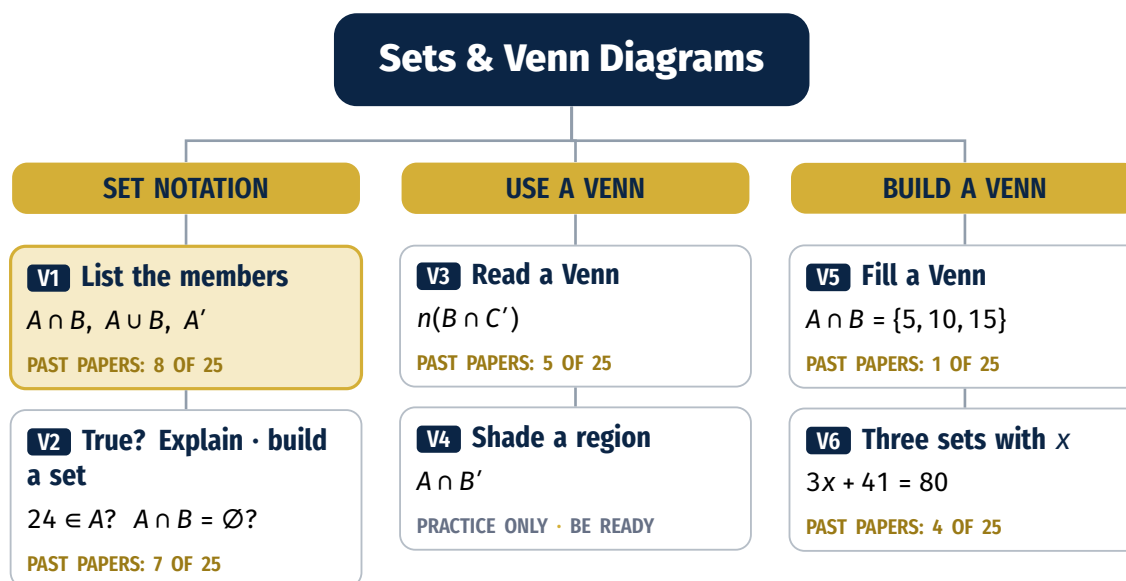
V1 List the members V2 True? Explain V3 Read a Venn

V4 Shade a region V5 Fill a Venn V6 Three sets with x

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Sets & Venn Diagrams at a glance



Your notes


DR ESLAM'S RULE

FILL FROM THE MIDDLE OUT

Centre first, then the overlaps, then the “only” parts, then outside.

 How often does each variant come up?

18 past-paper questions (Jan 2020 → Jun 2025) hold 25 set parts; 32 practice parts — grouped by type.

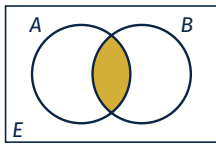
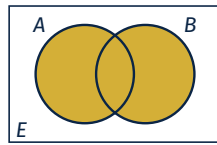
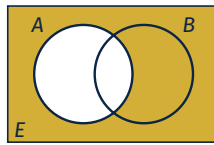
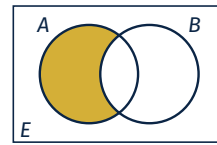
	PAST PAPERS (OUT OF 25)	PRACTICE PARTS
V1 List the members	8	10
V2 True? Explain • build a set	7	7
V3 Read a Venn	5	5
V4 Shade a region	0	2
V5 Fill a Venn	1	5
V6 Three sets with x	4	3

WHAT THIS TELLS YOU A set question is on most Paper 1s: listing $n, \cup, '$ (easy marks) plus a “reason” part. The 6-mark three-circle x question appears every year or so.

BEFORE YOU START

The symbols



Your notes
 $A \cap B$ "and" $A \cup B$ "or" A' "not A" $A \cap B'$ "A only"

E
universal set
Everything in the
question —
nothing outside
it.

$\in / \notin$
 $9 \in A, 24 \notin E$
"is a member of"
/ "is not".

$\emptyset$
 $A \cap C = \emptyset$
The empty set —
no members.

$n(A)$
 $n(A) = 6$
The **number** of
members.



Words to symbols

both / and

$\cap$
in A **and** in B

either / or

$\cup$
in A, in B, or in both

not / neither

$'$
everything outside



CALCULATOR FIRST, THEN CHECK

Write E out in full first, then each set in full: "multiples of 3" in $\{9, \dots, 20\}$ is $\{9, 12, 15, 18\}$.
Tick each member as you place it — every member of E goes in exactly one region.

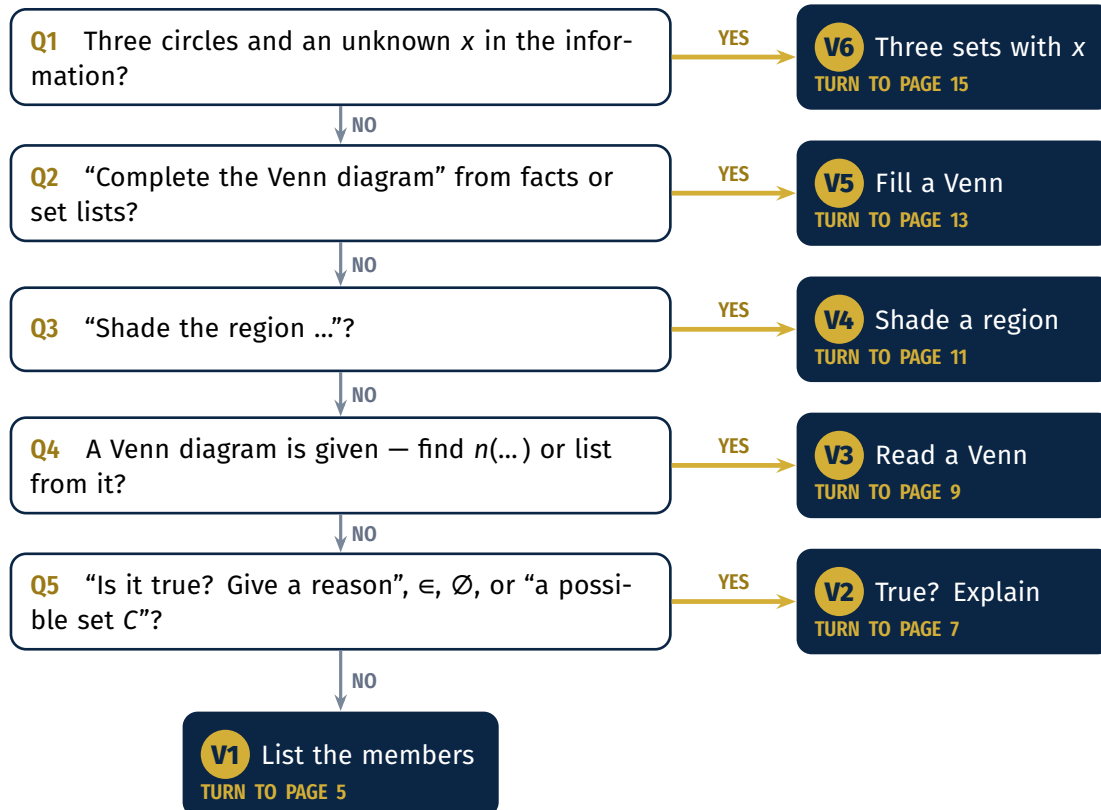
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1 MOST COMMON IN THE EXAM

List the Members

Write every set in full, then pick out $\cap$, $\cup$ or $'$.

SEEN IN PAST PAPERS

8 / 25

PRACTICE QUESTIONS: 10



Your notes

Came up in: Jan 2020 P1HR Q4 • Jan 2021 P1H Q4 • Jan 2021 P2HR Q3 • May 2021 P2H Q4 • Jun 2022 P2HR Q4 • May 2024 P1H Q7 • Jun 2024 P2HR Q4 • Jun 2025 P1HR Q5

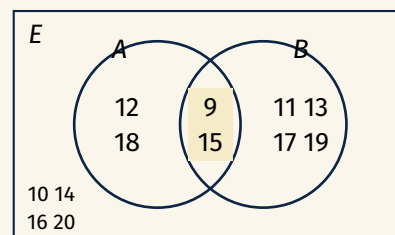
? What is it?

In plain words

Sets are described in words (“odd numbers”). First **write each set in full** — only numbers from E .

Then: $\cap$ = in **both**, $\cup$ = in **either** (no repeats), $'$ = in E but **not** in A .

IT LOOKS LIKE

 $A \cap B$ $A \cup B$ A' $W \cap G'$ 

How do I do it?

1 Write E

List every member of the universal set.

2 Write each set

Only members of E that fit the rule.

3 Pick out

 $\cap$ both • $\cup$ either • $'$ not.

4 Curly brackets

Answer as a set: $\{9, 15\}$ — each member once.

WRITE EVERY SET IN FULL FIRST Then it is just matching.

V1 List the Members

VARIANT 1 • SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO*"List the members of the set $A \cap B$ "*→ **Members in both***" $A \cup B$ "*→ **Members in either — write each once***" A' " or " $W \cap G'$ "*→ **Not in A / in W but not in G** **WORKED EXAMPLE****PAST PAPER**

June 2022 • Paper 2HR • Q4(a) • 2 marks

$E = \{9, 10, \dots, 20\}$, $A = \{\text{multiples of 3}\}$, $B = \{\text{odd numbers}\}$. List the members of (i) $A \cap B$ (ii) $A \cup B$.

STEP 2

$A = \{9, 12, 15, 18\}$

◀ multiples of 3 in E **STEP 2**

$B = \{9, 11, 13, 15, 17, 19\}$

◀ odd numbers in E **STEP 3**

$A \cap B = \{9, 15\}$

◀ in both

STEP 3

$A \cup B = \{9, 11, 12, 13, 15, 17, 18, 19\}$

◀ in either, once each

ANSWER (i) $\{9, 15\}$ (ii) $\{9, 11, 12, 13, 15, 17, 18, 19\}$ ✓ 8 members, none repeated

**WATCH OUT**✗ $A \cup B = \{9, 12, 15, 18, 9, 11, 13, 15, \dots\}$

✓ each member once

A set never repeats a member.

**EXAM TIP**

Use only numbers from E : 21 is a multiple of 3 but is not in $E = \{9, \dots, 20\}$.

**Check yourself**

calculator first • answers on page 17

1.1

Jan 2021 P1H

$E = \{20, \dots, 29\}$, A odd, B multiples of 3. List $A \cap B$ and $A \cup B$.

Answer:

1.2

Jun 2025 P1HR

$E = \{1, \dots, 12\}$,
 $A = \{2, 4, 6, 8, 10, 12\}$,
 $B = \{3, 6, 9, 12\}$. List
 $A \cup B$.

Answer:

1.3

Jun 2025 P1HR

With the sets in 1.2, list
 B' .

Answer:

V2

VARIANT 2

True? Explain • Build a Set

Decide with a reason that names a member
— or build a set that fits all the conditions.

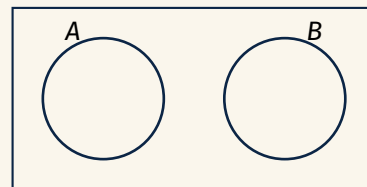
SEEN IN PAST PAPERS

7 / 25

PRACTICE QUESTIONS: 7


Your notes

Came up in: Jan 2020 P1HR Q4(b) • Jan 2021 P2HR Q3(b) • May 2021 P2H Q4(b) • Jun 2022 P2HR Q4(b)(c) • May 2024 P1H Q7(b)(c)
• Jun 2024 P2HR Q4(b)(c) • Jun 2025 P1HR Q5(b)

? What is it?**In plain words****Is it true?** Give the decision**and a member that proves it:** “No — 2 and 3 are in both A and B.”**Build a set C:** meet every condition one at a time; check at the end. $\emptyset$ means **no members**: $A \cap B = \emptyset$ means the circles do not overlap. $A \cap B = \emptyset$: **NO OVERLAP****IT LOOKS LIKE**

$A \cap B = \emptyset?$

$24 \in A?$

$C \cap B' = \{10$

$18\}$

How do I do it?**1 Write the sets**

In full, as in V1.

2 Test the claim

Look for a member that breaks (or proves) it.

3 Reason

Decision + the member: “No, 24 is not in E.”

4 Build a set

Choose members condition by condition; then check every condition.

A REASON NAMES A MEMBER

“Yes/No” alone scores 0.

V2 True? Explain · Build a Set

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

"Harpreet states that $A \cap B = \emptyset$.
Explain why she is incorrect"

→ Name a member in both

"Is it true that $24 \in A$?"

→ Check E first — is 24 even in it?

"List the members of one possible set
 C "

→ Build it condition by condition, then check

WORKED EXAMPLE **PAST PAPER**

June 2024 · Paper 2HR · Q4(b)(c) · 3 marks

$E = \{1, \dots, 10\}$, $A = \{\text{factors of } 6\}$, $B = \{\text{prime numbers}\}$. (b) Why is $A \cap B = \emptyset$ wrong? (c) C has 4 members; $A \cap C$ and $B \cap C$ have 2 members each and nothing in common. List C .

STEP 1 $A = \{1, 2, 3, 6\}$, $B = \{2, 3, 5, 7\}$ ◀ write the sets

STEP 3 2 and 3 are in both A and B , so $A \cap B \neq \emptyset$ ◀ (b) reason with members

STEP 4 $A \cap C = \{1, 6\}$, $B \cap C = \{5, 7\}$ ◀ (c) avoid 2, 3 (they are in both)

ANSWER (b) 2 and 3 are in both (c) $C = \{1, 5, 6, 7\}$ ✓ each condition checked

WATCH OUT

× "No, because they share numbers."

✓ "No — 2 and 3 are in both A and B ."

Name the members: that is the mark.

EXAM TIP

When you build a set, write the check line: $A \cap C = \{1, 6\}$, $B \cap C = \{5, 7\}$ — 2 members each ✓.

Check yourself

calculator first · answers on page 17

2.1 Jun 2022 P2HR

$E = \{9, \dots, 20\}$,
 $A = \{\text{multiples of } 3\}$. Is it
true that $24 \in A$?

Answer:

2.2 Jan 2020 P1HR

$B = \{b, l, u, e\}$,
 $G = \{g, r, e, y\}$,
 $W = \{w, h, i, t, e\}$. Is
 $B \cap G \cap W = \emptyset$?

Answer:

2.3 Jan 2021 P2HR

$E = \{21, \dots, 30\}$,
 $A = \{22, 24, 26, 28, 30\}$,
 $B = \{21, 24, 27, 30\}$,
 $C = \{23, 25, 29\}$. Write C
in terms of A and B .

Answer:

V3

VARIANT 3

Read a Venn Diagram

Numbers in regions: add the regions that the notation describes.

SEEN IN PAST PAPERS

5 / 25

PRACTICE QUESTIONS: 5



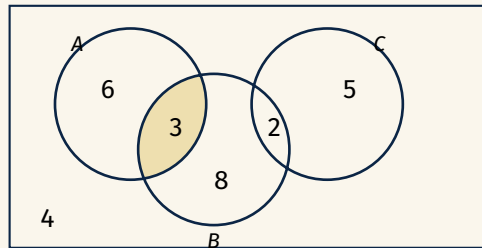
Your notes

Came up in: May 2021 P1H Q19 • Jan 2022 P1HR Q17 • May 2023 P1HR Q2 • Nov 2023 P1H Q1 • Nov 2024 P1H Q4

? What is it?**In plain words**

Each number is **how many** are in that region. $n(A \cup B)$: add **every region inside** A or B.

Shade it in your head (or lightly with pencil), then add the shaded numbers.

IT LOOKS LIKE $n(A \cup B)$ $n(A \cap C)$ $n(B \cap C')$ 

Jan 2022 P1HR Q17: A and C do not overlap.

How do I do it?**1 Read the notation**

$\cup$ either, $\cap$ both, ' outside.

2 Mark the regions

Lightly tick each region that fits.

3 Add

Add the numbers in the ticked regions.

4 Check the total

$n(E)$ is all regions, outside included.

SHADE, THEN ADD

Never add a region twice.

V3 Read a Venn Diagram

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“6, 3, 8 ... represent the numbers of elements”

→ **Add regions — do not list**

“Find $n(B \cap C')$ ”

→ **In B, not in C**

“ $n(A' \cup B' \cup C')$ ”

→ **Everything except the region in all three****WORKED EXAMPLE PAST PAPER**

January 2022 • Paper 1HR • Q17 • 4 marks

A only 6, $A \cap B$ 3, B only 8, $B \cap C$ 2, C only 5, outside 4 (A and C do not meet). Find (i) $n(A \cup B)$ (ii) $n(A \cap C)$ (iii) $n(B \cap C')$ (iv) $n(A' \cup B' \cup C')$.

STEP 3 $n(A \cup B) = 6 + 3 + 8 + 2 = 19$

◀ (i) every region in A or B

STEP 3 $n(A \cap C) = 0$

◀ (ii) the circles do not overlap

STEP 3 $n(B \cap C') = 3 + 8 = 11$

◀ (iii) in B, not in C

STEP 4 $n(A' \cup B' \cup C') = 28$

◀ (iv) no region is in all three: everything

ANSWER 19, 0, 11, 28✓ total $6 + 3 + 8 + 2 + 5 + 4 = 28$ **WATCH OUT**✗ $n(A \cup B) = 6 + 3 + 8 = 17$ ✓ $6 + 3 + 8 + 2 = 19$

The 2 is in B (and C) — it belongs to $A \cup B$ too.

EXAM TIP

Find $n(E)$ first — many “” answers are $n(E)$ minus something.

Check yourself

calculator first • answers on page 17

3.1

Dr Eslam

A only 7, $A \cap B$ 4, B only 9, outside 5. Find $n(A \cup B)$.

Answer:

3.2

Dr Eslam

Same diagram: find $n(B')$.

Answer:

3.3

Dr Eslam

Same diagram: find $n(A \cap B')$.

Answer:

V4

VARIANT 4

Shade a Region

Turn the notation into a region: $\cap$ is the overlap, $'$ is outside.

SEEN IN PAST PAPERS

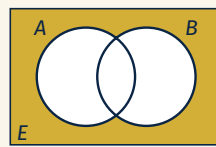
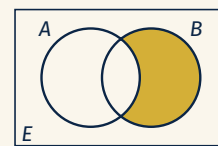
0 / 25

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 2


Your notes
? What is it?**In plain words**

Build the region in steps: shade the first part lightly, then keep only what the **next symbol** allows.

$A \cap B'$: inside A , outside B — the “**A only**” crescent.

IT LOOKS LIKE $A \cap B'$ $(A \cup B)'$ $A' \cap B$  $(A \cup B)'$  $A' \cap B$ **How do I do it?****1 Brackets first**

Work out the bracket region, e.g. $A \cup B$.

2 Apply $'$

Swap shaded and unshaded (everything else in E).

3 Apply $\cap$

Keep only regions shaded in **both** parts.

4 Check

Pick a point in your shading: does it fit the words?

$\cap$ = KEEP THE OVERLAP • $'$ = SWAP Build it one symbol at a time.

V4 Shade a Region

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

- “Shade the region $A \cap B'$ ” → **A only**
- “ $(A \cup B)'$ ” → **Outside both circles**
- “Describe the shaded region in set notation” → **Read it: in? out? → $\cap, '$**

WORKED EXAMPLE DR ESAM

Shading · 2 marks

On a two-set Venn diagram, shade the region $A' \cap B$.

- STEP 1** no brackets ◀ brackets first
- STEP 2** A' : everything outside circle A ◀ apply $'$
- STEP 3** keep only the part also in B ◀ apply $\cap$
- STEP 4** a point in “B only” is in B and not in A ✓ ◀ check

ANSWER **B only (the crescent of B outside A)** ✓ see the picture on page A

WATCH OUT

- ✗ shading all of B
- ✓ only the part of B outside A
- A' removes the overlap.

EXAM TIP

Shade clearly and only the region asked
— extra shading loses the mark.

Check yourself

calculator first · answers on page 17

4.1 Dr EslamWhich region is $A \cap B'$?

Answer:

4.2 Dr EslamWhich region is $(A \cup B)'$?

Answer:

4.3 Dr EslamWhich region is $(A' \cup B')'$?

Answer:

V5

VARIANT 5

Fill a Venn Diagram

Fill from the middle out: the overlap first, then the “only” parts, then outside.

SEEN IN PAST PAPERS

1 / 25

PRACTICE QUESTIONS: 5



Your notes

Came up in: Jan 2022 P1H Q7

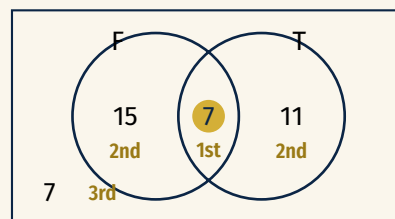
? What is it?**In plain words**

The overlap first — it belongs to both sets. Then each “only” part is the set **minus the overlap**. Then outside = total minus everything inside.

With numbers: 22 football, 7 both → football only $22 - 7 = 15$.

IT LOOKS LIKE

22 football, 18 tennis, 7 both

**How do I do it?****1 Overlap**

Put the “both” members (or number) in the middle.

2 Only parts

Each set minus what is already in the overlap.

3 Outside

What is left of E (or total minus inside).

4 Check

Every member placed once / regions add to the total.

FILL FROM THE MIDDLE OUT Dr Eslam's order.

V5 Fill a Venn Diagram

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Complete the Venn diagram for this information”

→ **Overlap, then only parts, then outside**

A' and B' given

→ **A only = B' minus A' (in A , not in B)**

WORKED EXAMPLE **PAST PAPER**

January 2022 · Paper 1H · Q7 · 3 marks

$E = \{4, \dots, 15\}$, $A \cap B = \{5, 10, 15\}$, $B' = \{7, 8, 9, 11, 12, 13, 14\}$, $A' = \{4, 6, 7, 8, 14\}$. Complete the Venn diagram.

STEP 1 middle: 5, 10, 15

◀ the overlap first

STEP 2 A only = $\{9, 11, 12, 13\}$, B only = $\{4, 6\}$

◀ A only: in B' not A' ; B only: in A' not B'

STEP 3 outside = $A' \cap B' = \{7, 8, 14\}$

◀ in neither

ANSWER A only 9,11,12,13 · both 5,10,15 · B only 4,6 · outside 7,8,14
members placed

✓ all 12

WATCH OUT

✗ writing 22 in the football-only region

✓ $22 - 7 = 15$

22 is the whole circle — the overlap is part of it.

EXAM TIP

Count at the end: the numbers in all regions must add up to $n(E)$.

Check yourself

calculator first · answers on page 17

5.1

Dr Eslam

40 students: 22 play football, 18 tennis, 7 both. How many play neither?

Answer:

5.2

Practice · Hard

25 people each have a cat or a dog or both; 17 have a cat, 15 a dog. How many have both? Only one?

Answer:

5.3

Practice · Easy

50 people: 30 have a cat, 25 a dog, 11 both. How many have neither?

Answer:

V6

VARIANT 6

Three Sets with x

Three circles, one unknown: fill from the centre, write the total as an equation in x .

SEEN IN PAST PAPERS

4 / 25

PRACTICE QUESTIONS: 3



Your notes

Came up in: Jan 2020 P2HR Q17 • Jan 2021 P2H Q16 • May 2023 P1H Q16 • Nov 2024 P2H Q21

? What is it?**In plain words**

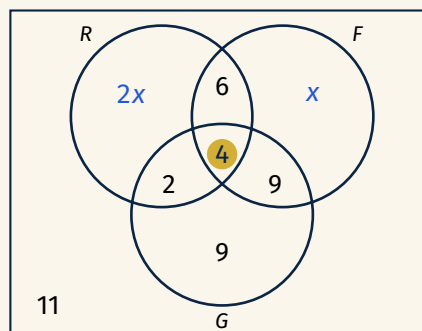
Fill the **centre** (all three) first, then each **pair-only** region (pair total minus the centre), then each **“only”** region. Some regions are in terms of x .

The regions **add up to the total**: that is the equation for x .

IT LOOKS LIKE

4 study all three

Russian only is twice French only



Jan 2021 P2H Q16, filled from the centre.

How do I do it?**1 Centre**

The number in all three.

2 Pair-only

Each “both A and B” total minus the centre.

3 Only regions

Set total minus its other three regions — or the x expressions.

4 Equation

All regions + outside = total. Solve, then answer the question.

ALL REGIONS = THE TOTAL That is your equation in x .

V6 Three Sets with x

VARIANT 6 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“10 study Russian and French”
(includes all three)

$$\rightarrow \text{Pair-only} = 10 - 4 = 6$$

“the number who study Russian only is
twice French only”

$$\rightarrow F \text{ only} = x, R \text{ only} = 2x$$

“Given that a total of 80 ...”

$$\rightarrow \text{Sum of all regions} = 80$$

WORKED EXAMPLE **PAST PAPER**

January 2021 · Paper 2H · Q16 · 6 marks

All three 4; R&F 10; F&G 13; R&G 6; G 24; none 11; Russian only is twice French only (x). 80 students. How many study Russian?

STEP 1 centre 4 ◀ all three

STEP 2 R&F only 6, F&G only 9, R&G only 2 ◀ pair totals - 4

STEP 3 G only = $24 - 4 - 9 - 2 = 9$, R only = $2x$ ◀ only regions

STEP 4 $2x + x + 6 + 4 + 2 + 9 + 9 + 11 = 80 \Rightarrow$ ◀ all regions = total
 $3x + 41 = 80 \Rightarrow x = 13$

ANSWER Russian = $2(13) + 6 + 4 + 2 = 38$

✓ regions add to 80 ✓

WATCH OUT

✗ putting 10 in the R&F region

✓ $10 - 4 = 6$

“Russian and French” counts the 4 who do all three too.

EXAM TIP

6 marks: diagram (3), equation (1), x (1), the final answer (1). Finish the question — here, Russian.

Check yourself

calculator first · answers on page 17

6.1 Jan 2020 P2HR

50 asked. D 28, C 18, R 20, C&R 8, D&R 9, D&C x , all three 6, none 5. Find x and $n(C' \cap D')$.

Answer:

6.2 Dr Eslam

60 people: all three 4; pair-only 6, 5, 3; A only x , B only $2x$, C only 10; none 8. Find x .

Answer:

6.3 Dr Eslam

Centre 2; each pair-only 3; A only x , B only x , C only $2x$; none 4; total 39. Find x .

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with Python set operations and sympy (check_answers.py).

1.1 {21, 27};
{21, 23, 24, 25, 27, 29}

1.2
{2, 3, 4, 6, 8, 9, 10, 12}

1.3
{1, 2, 4, 5, 7, 8, 10, 11}

2.1 No: 24 is not in E

2.2 No: e is in all three

2.3 $C = (A \cup B)'$

3.1 20

3.2 12

3.3 7

4.1 A only

4.2 outside both
circles

4.3 $A \cap B$ (the overlap)

5.1 7

5.2 both 7, only one 18

5.3 6

6.1 $x = 10$;
 $n(C' \cap D') = 14$

6.2 $x = 8$

6.3 $x = 6$

THE WHOLE TOPIC ON ONE PAGE

Sets & Venn Diagrams cheat sheet



Your notes
V1 List the members 8/25

$$A \cap B = \{9, 15\}$$

Write every set in full first.

V2 True? Explain 7/25

$$2, 3 \in A \cap B$$

Decision + a named member.

V3 Read a Venn 5/25

$$n(A \cup B) = 6 + 3 + 8 + 2$$

Shade in your head, then add.

V4 Shade a region

$$A \cap B' = A \text{ only}$$

 n keep overlap • ' swap.**V5 Fill a Venn 1/25**

$$22 - 7 = 15$$

Middle first, then only parts, then outside.

V6 Three sets 4/25

$$3x + 41 = 80$$

Centre $\rightarrow$ pairs $\rightarrow$ onlys; regions = total.**TOP 3 LOST MARKS**

1. a member not in $E \times$
2. a pair total put in the pair-only region $\times$
3. a reason with no member named $\times$

BOTTOM LINE n both • $\cup$ either • ' not • fill from the middle out • regions add to the total.